

Control-Lyapunov Functions for Time-Varying Set Stabilization*

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Abstract. This paper shows that, for time varying systems, global asymptotic controllability to a given closed subset of the state space is equivalent to the existence of a continuous control-Lyapunov function with respect to the set.

Keywords. Nonlinear control, Stability, Control Lyapunov function.

1 Introduction

We will study continuous-time systems with dynamics given by differential equations of the type:

$$\dot{x}(t) = f(t, x(t), u(t)), \quad (1)$$

where $x(t) \in \mathbb{R}^n$ represents the state variable and $u(t) \in U$ is the control variable. (Technical assumptions on f and U are described below.)

We are interested in questions of stabilization relative to a subset $\mathcal{A}$ of the state space $\mathbb{R}^n$. For example, the set $\mathcal{A}$ may be just an equilibrium point, or it may represent a target subset of a different kind. This target set might be a desired periodic orbit, or, in the context of designing observers, the Equations (1) might represent a composite state $x = (x_1, x_2)$, consisting of the state x_1 of the original system together with the state x_2 of an observer; in that case, $\mathcal{A}$ would be the set of states x for which identification has been achieved, that is, the set consisting of those x for which $x_1 = x_2$. (Note that in this last example, the set $\mathcal{A}$ is not bounded.) There are several motivations for considering time-varying dynamics. For instance, in this manner one may encompass problems of tracking, in which the difference between some variables and a signal to be followed evolves according to a differential equation which depends explicitly in t . The purpose of this paper is to generalize the results of the paper [8], which dealt only with $\mathcal{A} = \{0\}$ and time-invariant systems, to the general model (1).

A classical technique for stabilization is to look for Lyapunov-type functions, which play the role of abstract “energy” or “cost” functions that can be made to decrease in directions corresponding to possible controls, as long as the state is away from $\mathcal{A}$. For smooth such functions V , and taking for simplicity the case when V and f are time-invariant, and $\mathcal{A} = \{0\}$, this amounts to the requirement that, for each nonzero state x , there must be some control value $u = u_x$ so that $\nabla V(x) \cdot f(x, u) < 0$. One uses the generic term

“control-Lyapunov function” (clf) for such a function. The clf approach, assuming a V has been found, allows the search for stabilizing inputs by iteratively solving a static nonlinear programming problem: when at state x , find u such that this inequality holds. Recent expositions of results about (smooth) clf’s can be found in the textbooks [5] and [6]. An obvious question is whether the existence of a continuously differentiable clf is equivalent to the possibility of driving every state asymptotically to the set $\mathcal{A}$ (zero in this particular case). The answer is negative; for instance, if controls are in $\mathbb{R}^m$, and $f(x, u) = f_0 + \sum_{i=1}^m u_i f_i(x)$ is affine in u , the existence of a (smooth) V would imply that there is some feedback law $u=k(x)$ so that the origin is a globally asymptotically stable state for the closed-loop system $\dot{x} = f(x, k(x))$ and k is continuous on $\mathbb{R}^n \setminus \{0\}$. This was proved by Artstein in [2]. But continuous feedback may fail to exist, even for very simple controllable systems (see e.g. [9], Section 4.8). Thus smooth clf’s do not always exist, even if the system is asymptotically controllable.

However, it was shown in [8] that *continuous* clf’s do exist. Of course, one must modify the statement of the clf condition, since the gradient is not well-defined if V is not differentiable. This modification can be done in various ways. The first possibility is to ask that for each state $x \neq 0$ (or rather, for each state not in $\mathcal{A}$, in the general case) there exist a trajectory which is defined on a small interval of time and which decreases the value of V . This is the modification that was used in [8] (for time-invariant systems and the special case $\mathcal{A} = \{0\}$), and which we also employ in this paper. A second possibility, much more attractive, is to use generalized gradients of various types. We refer the reader to [10] and [4] for descriptions of this second approach, including, in the second citation, an application of these ideas to the construction of discontinuous stabilizing feedback. The two approaches are equivalent, but providing technical details is impossible in this short paper, due to space restrictions; thus, we limit ourselves to the first possibility, and leave the equivalence for a fuller journal version.

The proofs are in general based on the ideas in [8], but when dealing with time varying systems, and espe-

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cially with possibly non-compact sets $\mathcal{A}$, many technical complications arise. (Of course, one must allow now for time-dependent V , and the definition of asymptotic controllability must be in some sense uniform on time, in order to obtain a necessary and sufficient result.) We also use some ideas from the paper [7], which dealt with set stability but only for systems with no controls (which allows constructing *smooth* V 's).

Definitions and Main Result

We take U to be a locally compact metric space, with a distinguished "0" element. We denote $U_r = \{u \mid d(u, 0) \leq r\}$. The map f is supposed to be measurable in t , and locally Lipschitz in (x, u) uniformly for t in a bounded interval. The set of control maps $\mathbf{u} \in \mathbf{U}$ are measurable essentially bounded functions $\mathbf{u} : [t_0, \infty) \rightarrow U$, with $t_0 \in \mathbb{R}$. We denote by $\|\mathbf{u}\|$ the essential supremum of the map $\mathbf{u}$. Notice that with these assumptions we guarantee the local existence and uniqueness of solutions of (1). For a given $(t_0, x_0) \in \mathbb{R} \times \mathbb{R}^n$, and a control map $\mathbf{u} : [t_0, \infty)$, we denote by $x(t; t_0, x_0, \mathbf{u})$ the solution of (1) with initial condition $x(t_0, t_0, x_0, \mathbf{u}) = x_0$. In general this solution will be defined on an interval of the form $[t_0, \bar{t})$.

A function $\gamma : \mathbb{R}_{\geq 0} \rightarrow \mathbb{R}_{\geq 0}$ is called a $\mathcal{K}$ -function if it is continuous, strictly increasing, and $\gamma(0) = 0$; a $\mathcal{K}$ -map γ is called a $\mathcal{K}_\infty$ -function if $\lim_{s \rightarrow \infty} \gamma(s) = \infty$. A function $\beta : \mathbb{R}_{\geq 0} \times \mathbb{R}_{\geq 0} \rightarrow \mathbb{R}_{\geq 0}$ is said to be a $\mathcal{KL}$ -function if for each fixed $t \geq 0$ the map $\beta(\cdot, t)$ is a $\mathcal{K}$ -function, and if for each fixed s the map $\beta(s, \cdot)$ is decreasing to zero as $t \rightarrow \infty$. Given a non-empty closed set $\mathcal{A} \subset \mathbb{R}^n$, we denote by $|x|_{\mathcal{A}}$ the distance from x to $\mathcal{A}$, and we say that $\mathcal{A}$ is *weakly invariant* if, for all $x_0 \in \mathcal{A}$ and all $t_0 \in \mathbb{R}$, there exists a control map $\mathbf{u}_0$ such that $x(t; t_0, x_0, \mathbf{u}_0)$ is defined and lies in $\mathcal{A}$ for all $t \geq t_0$.

Definition 1.1 Let $\mathcal{A} \subset \mathbb{R}^n$ be a closed, weakly invariant, and nonempty set. We say that (1) is *globally asymptotically controllable* (GAC) to $\mathcal{A}$ if there exist a $\mathcal{KL}$ -function $\beta(\cdot, \cdot)$, and a continuous, positive and increasing function $\gamma(\cdot)$ such that: for each $(t_0, x_0) \in \mathbb{R} \times \mathbb{R}^n$ there exists a control function $\mathbf{u} : [t_0, \infty) \rightarrow U$, with $\|\mathbf{u}\| \leq \gamma(|x_0|_{\mathcal{A}})$, such that the corresponding trajectory $x(t) = x(t, t_0, x_0, \mathbf{u})$ exists for all $t \geq t_0$ and satisfies:

$$|x(t)|_{\mathcal{A}} \leq \beta(|x_0|_{\mathcal{A}}, t - t_0) \quad \text{for all } t \geq t_0. \quad (2)$$

An alternative way to define GAC to a set $\mathcal{A}$ is as follows. The proof, which we omit for reasons of space, consists of a routine argument involving comparison functions, and is patterned along the lines of the completely analogous result in [7], Proposition 2.5.

Lemma 1.2 The system (1) is GAC to $\mathcal{A}$ if and only if there exists a continuous, positive and increasing function γ_1 such that:

1. there exists a $\mathcal{K}_\infty$ -map γ_2 such that for all $\epsilon > 0$, if $|x_0|_{\mathcal{A}} \leq \gamma_2(\epsilon)$ and $t_0 \in \mathbb{R}$, then there exist $t_1 > t_0$, and $\mathbf{u} : [t_0, t_1) \rightarrow U$, with $\|\mathbf{u}\| \leq \gamma_1(|x_0|_{\mathcal{A}})$ such that $|x(t, t_0, x_0, \mathbf{u})|_{\mathcal{A}} \leq \epsilon$ for all $t \geq t_0$;

2. for any r, ϵ there exists $T \in \mathbb{R}$ such that given $(t_0, x_0) \in \mathbb{R} \times \mathbb{R}^n$, with $|x_0|_{\mathcal{A}} \leq r$, then there exists $\mathbf{u} : [t_0, \infty) \rightarrow U$ with $\|\mathbf{u}\| \leq \gamma_1(|x_0|_{\mathcal{A}})$ such that $|x(t, t_0, x_0, \mathbf{u})|_{\mathcal{A}} \leq \epsilon$ for all $t \geq T + t_0$

We now recall some standard notions regarding relaxed controls. The set of relaxed controls $\mathbf{W}$ is the set of measurable functions $\mathbf{w} : [t_0, \infty) \rightarrow P(U)$ where $P(U)$ is the set of probability measures on U equipped with the weak topology. We let $\mathbf{W}_r$ denote the set of those relaxed controls $\mathbf{w}$ such that $\mathbf{w}(t)$ has support on U_r for almost all t . An ordinary control $\mathbf{u} \in \mathbf{U}$ can be seen as a relaxed one if $\mathbf{u}(t)$ is identified with the Dirac measure concentrated in $\mathbf{u}(t)$. One defines a topology on $\mathbf{W}$ characterized by weak convergence: $\mathbf{w}_k \rightarrow \mathbf{w}$ if and only if $\int_T \int_U g(t, u) d\mathbf{w}_k(u) dt \rightarrow \int_T \int_U g(t, u) d\mathbf{w}(u) dt$ for all functions $g : T \times U \rightarrow \mathbb{R}$ which are continuous in u , measurable in t , and such that $\max\{|g(t, u)|, u \in U\}$ is integrable on T , where T is a bounded interval. With this topology $\mathbf{U}_r$ is dense in $\mathbf{W}_r$, and $\mathbf{W}_r$ is sequentially compact. We let $\|\mathbf{w}\| = \inf\{r \mid \mathbf{w}(t) \in P(U_r) \text{ for a.e. } t\}$ (notice that for ordinary controls this is the essential supremum). Given $w \in P(U_r)$, one extends f to relaxed controls by defining $f(t, x, w) := \int_{U_r} f(t, x, s) dw(s)$. As for ordinary controls, given $(t_0, x_0) \in \mathbb{R} \times \mathbb{R}^n$, and a relaxed control $\mathbf{w} : [t_0, \infty) \rightarrow P(U)$, we denote by $x(t; t_0, x_0, \mathbf{w})$ the solution of (1) with initial condition $x(t_0; t_0, x_0, \mathbf{w}) = x_0$. For more details on relaxed controls see [1].

Definition 1.3 A continuous function $V : \mathbb{R} \times \mathbb{R}^n \rightarrow \mathbb{R}_{\geq 0}$ is a *Lyapunov function* for the model (1) with respect to the closed, weakly invariant, and non-empty set $\mathcal{A}$ if the two following properties hold.

1. There exist two $\mathcal{K}_\infty$ -maps α_1, α_2 such that, for all $(t, x) \in \mathbb{R} \times \mathbb{R}^n$: $\alpha_1(|x|_{\mathcal{A}}) \leq V(t, x) \leq \alpha_2(|x|_{\mathcal{A}})$.
2. There exist a continuous, positive and increasing map ϕ , and a $\mathcal{K}$ -function α_3 , such that for each $(t_0, x_0) \in \mathbb{R} \times \mathbb{R}^n$, there exist $t_1 > t_0$, and $\mathbf{w} : [t_0, t_1) \rightarrow P(U)$ with $\|\mathbf{w}\| \leq \phi(|x_0|_{\mathcal{A}})$ and: $\liminf_{t \rightarrow t_0^+} \frac{V(t, x(t)) - V(t_0, x_0)}{t - t_0} \leq -\alpha_3(|x_0|_{\mathcal{A}})$, where $x(t) = x(t, t_0, x_0, \mathbf{w})$.

Remark 1.4 From the proof of Theorem 1 it will be clear that in the definition of Lyapunov function, we could have used the $\limsup$ instead of the $\liminf$, and the same result will hold. In that case, however, the sufficiency part of the result becomes weaker.

Definition 1.5 Given a closed and non-empty set $\mathcal{A}$, we say that the function $f(t, x, u)$ satisfies the *boundedness assumption* with respect to $\mathcal{A}$ if it holds that, for each $r_1, r_2 > 0$:

$$\sup_{\substack{|x|_{\mathcal{A}} \leq r_1 \\ d(u, 0) \leq r_2}} |f(t, x, u)| \leq M_{r_1, r_2} \quad \text{a.e. } t \in \mathbb{R}. \quad (3)$$

Notice that if f is independent of t and continuous and $\mathcal{A}$ is compact then (3) holds.

Theorem 1 Let Σ be a given model of type (1), and $\mathcal{A} \subset \mathbb{R}^n$ be a closed, weakly invariant, and non-empty set. Assume that the model satisfies the boundedness assumption with respect to $\mathcal{A}$ (Definition 1.5). Then (1) is GAC to $\mathcal{A}$ if and only if there exists a Lyapunov function V for (1) with respect to $\mathcal{A}$.

The sufficiency part is much easier to establish than the necessity part. One way to prove that the existence of a Lyapunov function is enough to guarantee GAC is by adapting the arguments used in [8] to this setting. This needs some attention, since we are in a non-autonomous case and the set $\mathcal{A}$ is possibly not compact.

An alternative way to prove the sufficiency result is through the use of recent results on set-valued maps and non-smooth analysis (see [3]). By using Theorem 2.1 (in the non-autonomous setting) and Theorem 7.1 of [3] one first proves that if V is a Lyapunov function for (1) with respect to the set $\mathcal{A}$, then for each $(t_0, x_0) \in \mathbb{R} \times \mathbb{R}^n$, there exists a relaxed control $\mathbf{w}_0$ such that $\|\mathbf{w}_0\| \leq \delta(|x_0|_{\mathcal{A}})$, and the corresponding trajectory, $x(t) = x(t; t_0, x_0, \mathbf{w}_0)$, exists for all $t \geq t_0$ and satisfies: $V(t, x(t)) - V(t_0, x_0) \leq -\int_{t_0}^t \alpha_3(|x(s)|_{\mathcal{A}}) ds$. Then, by using the previous equation, it is easy to see that properties 1. and 2. of Lemma 1.2 follow, and so one concludes that the model is GAC to $\mathcal{A}$.

The next section is devoted to prove the necessity part of Theorem 1.

2 Lyapunov Function Construction

The next proposition establishes a technical property of $\mathcal{KL}$ -functions.

Proposition 2.1 Let $\beta(\cdot, \cdot)$ be $\mathcal{KL}$ -function, and choose a strictly increasing sequence of positive real numbers, $\{\epsilon_i\}_{i \geq 0}$, such that $\lim_{i \rightarrow +\infty} \epsilon_i = +\infty$. Then there exists a continuous, strictly decreasing function $g : \mathbb{R}_{\geq 0} \rightarrow \mathbb{R}_{\geq 0}$, and a strictly increasing sequence $\{T_i\}_{i \geq 0}$, with $T_0 = 0$, such that: [i)] $\lim_{t \rightarrow +\infty} g(t) = 0$; [ii)] $\lim_{i \rightarrow +\infty} T_i = +\infty$; [iii)] if $p < \epsilon_i$ then $\beta(p, t) \leq g(t)$ for all $t \geq T_i$.

Proof. Let $T_0 = 0$ and, for $i \geq 1$, define, inductively on i :

$$\alpha_i = \beta(\epsilon_{i-1}, T_{i-1} + 1/i),$$

and

$$T_i \text{ such that } \beta(\epsilon_i, T_i) = \alpha_i/2 = g_i.$$

Notice that, since $\beta(\epsilon_i, T_{i-1} + 1/i) > \alpha_i$, it holds that:

$$T_i > T_{i-1} + 1/i \Rightarrow T_i > \sum_{j=1}^i 1/j, \quad \text{for } i \geq 1.$$

Thus the sequence $\{T_i\}_{i \geq 0}$ is strictly increasing and satisfies ii). Moreover, since: $\alpha_{i+1} = \beta(\epsilon_i, T_i + \frac{1}{i+1}) < \beta(\epsilon_i, T_i) = \frac{\alpha_i}{2}$, we have: $\alpha_{i+1} \leq \frac{\alpha_i}{2} \forall i \geq 1$, and thus

$$\lim_{i \rightarrow +\infty} \alpha_i = 0. \quad (4)$$

Let g_{-1} and g_0 be two constants such that $g_{-1} > g_0 > 2g_1$, and, for $i \geq 0$, let P_i be the point (T_i, g_{i-1}) . Let $l_{P_i P_{i+1}}(\cdot)$ be the linear function such $l_{P_i P_{i+1}}(T_i) = g_{i-1}$, and $l_{P_i P_{i+1}}(T_{i+1}) = g_i$. It is easy to see that, by choosing g_{-1} sufficiently large, we may assume:

$$\beta(x, t) < l_{P_0 P_1}(t) \forall t \in [T_0, T_1], \forall x < \epsilon_0. \quad (5)$$

Now let: $g(t) = l_{P_i P_{i+1}}(t)$ for $t \in [T_i, T_{i+1}]$. Then clearly the map $g(\cdot)$ is continuous and strictly decreasing. Moreover, by equation (4), we have $\lim_{t \rightarrow +\infty} g(t) = 0$ (thus i) holds). Now we establish iii). Let $p \in \mathbb{R}$, and assume $p < \epsilon_i$. If $i = 0$ and $t \in [T_0, T_1]$, then property iii) holds by construction (equation (5)). So we may assume $t \geq T_1$. Let $t \geq T_i$, then $t \in [T_j, T_{j+1}]$ with $j \geq i \geq 1$. Then we have: $g(t) \geq g_j = \alpha_j/2 = \beta(\epsilon_j, T_j)$. On the other hand, if $x < \epsilon_i$ and $t \in [T_j, T_{j+1}]$, we also have: $\beta(x, t) < \beta(\epsilon_i, t) \leq \beta(\epsilon_j, t) \leq \beta(\epsilon_j, T_j) = g_j \leq g(t)$; so iii) holds.

From now on, we assume given a system Σ and a closed, weakly invariant, and non-empty set $\mathcal{A} \subset \mathbb{R}^n$. Moreover we assume that Σ satisfies the boundedness assumption with respect to $\mathcal{A}$ (Definition (1.5)), and it is GAC to $\mathcal{A}$. Our aim is to construct a Lyapunov function for Σ with respect to the set $\mathcal{A}$. The idea of the construction is similar to the one given in [8].

Let $\gamma(\cdot)$ and $\beta(\cdot, \cdot)$ be respectively the continuous, positive and increasing, and $\mathcal{KL}$ maps given by Definition 1.1. Choose any $\epsilon_0 > 0$, and let $\{\epsilon_i\}_{i \geq 0}$ be any strictly increasing sequence such that:

$$\lim_{i \rightarrow +\infty} \epsilon_i = +\infty \quad \text{and} \quad \epsilon_{i+1} > \min\{\beta(\epsilon_i, 0), \gamma(\epsilon_i)\}.$$

Now let $\{T_i\}_{i \geq 0}$, and $g(\cdot)$ be the sequence and the function given by Proposition 2.1. Since the map g is strictly decreasing, and $g : \mathbb{R}_{\geq 0} \rightarrow (0, g(0)]$, we may let:

$$h(p) = \begin{cases} g^{-1}(p) & \text{if } p \in (0, g(0)], \\ 0 & \text{if } p > 0. \end{cases}$$

Notice that $h : \mathbb{R}_{>0} \rightarrow \mathbb{R}_{\geq 0}$ is decreasing, continuous, and $\lim_{p \rightarrow 0^+} h(p) = +\infty$. Now we define:

$$N(p) = \begin{cases} 0 & \text{if } p = 0, \\ p \exp\{-h(p)\} & \text{if } p > 0. \end{cases} \quad (6)$$

Notice that: [i)] N is a continuous and strictly increasing map; [ii)] $\lim_{p \rightarrow +\infty} N(p) = +\infty$; [iii)] if $p \leq \alpha$ then $N(p) \leq \alpha$. For each $i \geq 0$, let $M_i = M(\epsilon_{i+2}, \epsilon_{i+2})$, where $M(\epsilon_{i+2}, \epsilon_{i+2})$ are the constants given by the boundedness assumption, i.e.:

$$\sup_{\substack{|x|_{\mathcal{A}} \leq \epsilon_{i+2} \\ d(u, 0) \leq \epsilon_{i+2}}} |f(t, x, u)| \leq M_i \quad \text{for a.e. } t \in \mathbb{R}.$$

Clearly $M_i \leq M_{i+1}$.

Now let $M : \mathbb{R}_{\geq 0} \rightarrow \mathbb{R}_{\geq 0}$ and $\delta : \mathbb{R}_{\geq 0} \rightarrow \mathbb{R}_{\geq 0}$ be any two continuous and increasing maps such that:

$$M(p) \geq \begin{cases} M_0 & \text{if } p < \frac{\epsilon_0}{2}, \\ M_i & \text{if } \frac{\epsilon_{i-1}}{2} \leq p < \frac{\epsilon_i}{2}, \end{cases} \quad (7)$$

$$\delta(p) \geq \gamma(p) \quad \text{for all } p \geq 0, \quad (8)$$

and, for all $i \geq 0$,

$$\delta(\epsilon_{i+2}) \geq \frac{\epsilon_i}{2} N\left(\frac{\epsilon_i}{2}\right) + \delta(\gamma(\epsilon_0)). \quad (9)$$

Moreover, for $i \geq 1$, we require that, if $\epsilon_{i+1} < p \leq \epsilon_{i+2}$, then:

$$\delta(p) > \delta(\gamma(\epsilon_i)) + \quad (10)$$

$$M(\beta(\epsilon_i, 0)) [N(\beta(\epsilon_i, 0))T_i + \beta(\epsilon_i, 0)]$$

and, for $i \geq 0$:

$$M(\epsilon_{i+1}) > \left(\frac{2M_i}{N(\epsilon_{i+1})} \frac{1}{\epsilon_{i+2} - \epsilon_{i+1}} \right) \left\{ M(\beta(\epsilon_i, 0)) [N(\beta(\epsilon_i, 0))T_i + \beta(\epsilon_i, 0)] + \delta(\gamma(\epsilon_i)) \right\}. \quad (11)$$

Notice that, since $\min\{\beta(\epsilon_i, 0), \gamma(\epsilon_i)\} < \epsilon_{i+1}$, one shows easily that two continuous and increasing functions satisfying inequalities (7), (8), (9), (10), and (11) exist.

Finally, we let: $F(p) = M(p)N(p) = M(p)p \exp\{-h(p)\}$. Fix $(t_0, x_0) \in \mathbb{R} \times \mathbb{R}^n$. Then let ϵ_{i_0} be the first index such that $|x_0|_{\mathcal{A}} < \epsilon_{i_0}$. For each control $\mathbf{w}$, denote by $x(t) = x(t; t_0, x_0, \mathbf{w})$ and let:

$$Q(t_0, x_0, \mathbf{w}) = \int_{t_0}^{+\infty} F(|x(t)|_{\mathcal{A}}) dt + \max\{\delta(\|\mathbf{w}\|) - \delta(\gamma(\epsilon_0)), 0\}, \quad (12)$$

if $x(t)$ exists for all $t \geq t_0$ and is such that $|x(t)|_{\mathcal{A}} \leq \epsilon_{i_0+2}$ for all $t \geq t_0$. In all the other cases we let $Q(t_0, x_0, \mathbf{w}) = +\infty$. Then we let $V(t_0, x_0) = \inf_{\mathbf{w} \in \mathbf{W}} Q(t_0, x_0, \mathbf{w})$. We want to prove that the function V defined above is a Lyapunov function for the model (1) with respect to the set $\mathcal{A}$.

Lemma 2.2 Let $(t_0, x_0) \in \mathbb{R} \times \mathbb{R}^n$, and $i \geq 0$ be the first index such that $|x_0|_{\mathcal{A}} < \epsilon_i$. Then there exists an ordinary control $\mathbf{u}$, with $\|\mathbf{u}\| \leq \gamma(|x_0|_{\mathcal{A}})$, such that:

$$Q(t_0, x_0, \mathbf{u}) \leq F(\beta(|x_0|_{\mathcal{A}}, 0))T_i + M(\beta(|x_0|_{\mathcal{A}}, 0))\beta(|x_0|_{\mathcal{A}}, 0) + \max\{\delta(\gamma(|x_0|_{\mathcal{A}})) - \delta(\gamma(\epsilon_0)), 0\}. \quad (13)$$

Proof. Given $(t_0, x_0) \in \mathbb{R} \times \mathbb{R}^n$ with $|x_0|_{\mathcal{A}} < \epsilon_i$, let $\mathbf{u}$ be the control function given by the GAC property. Then $\|\mathbf{u}\| \leq \gamma(|x_0|_{\mathcal{A}})$, moreover the corresponding trajectory $x(t)$ is defined for all $t \geq t_0$ and satisfies $|x(t)|_{\mathcal{A}} \leq \beta(|x_0|_{\mathcal{A}}, t - t_0) < \beta(\epsilon_i, 0) < \epsilon_{i+2}$. Moreover, for $t \geq T_i + t_0$, we have: $|x(t)|_{\mathcal{A}} \leq \beta(|x_0|_{\mathcal{A}}, t - t_0) < \beta(\epsilon_i, t - t_0) < g(t - t_0)$. Thus, for all $t \geq T_i + t_0$, $h(|x(t)|_{\mathcal{A}}) \geq h(g(t - t_0)) = t - t_0$. So we have: $Q(t_0, x_0, \mathbf{u}) = \int_{t_0}^{t_0+T_i} F(|x(t)|_{\mathcal{A}}) dt + \int_{t_0+T_i}^{+\infty} F(|x(t)|_{\mathcal{A}}) dt + \max\{\delta(\|\mathbf{u}\|) - \delta(\gamma(\epsilon_0)), 0\} \leq F(\beta(|x_0|_{\mathcal{A}}, 0))T_i + M(\beta(|x_0|_{\mathcal{A}}, 0))\beta(|x_0|_{\mathcal{A}}, 0) + \max\{\delta(\gamma(|x_0|_{\mathcal{A}})) - \delta(\gamma(\epsilon_0)), 0\}$. Thus the Lemma is proved.

Remark 2.3 Notice that, from the previous lemma, we have in particular:

(a) if $|x_0|_{\mathcal{A}} < \epsilon_0$, then $T_0 = 0$ and the max is also zero, so: $V(t_0, x_0) \leq M(\beta(|x_0|_{\mathcal{A}}, 0))\beta(|x_0|_{\mathcal{A}}, 0)$,

(b) if $\epsilon_{i-1} \leq |x_0|_{\mathcal{A}} < \epsilon_i$, with $i \geq 1$, then

$$V(t_0, x_0) \leq \delta(\gamma(\epsilon_i)) - \delta(\gamma(\epsilon_0)) + M(\beta(\epsilon_i, 0)) [N(\beta(\epsilon_i, 0))T_i + \beta(\epsilon_i, 0)].$$

Lemma 2.4 There exists a $\mathcal{K}_\infty$ -map $\alpha_2(\cdot)$ such that, for all $(t_0, x_0) \in \mathbb{R} \times \mathbb{R}^n$, it holds that: $V(t_0, x_0) \leq \alpha_2(|x_0|_{\mathcal{A}})$.

Proof. For $\epsilon_{i-1} \leq p < \epsilon_i$ ($\epsilon_{-1} = 0$), let

$$\tilde{\alpha}_2(p) = F(\beta(p, 0))T_i + M(\beta(p, 0))\beta(p, 0) + \max\{\delta(\gamma(p)) - \delta(\gamma(\epsilon_0)), 0\}.$$

Then $\tilde{\alpha}_2(\cdot)$ is increasing and $\tilde{\alpha}_2(0) = 0$. Now let $\alpha_2(\cdot)$ be any $\mathcal{K}_\infty$ -function such that $\tilde{\alpha}_2(p) \leq \alpha_2(p)$. For each $(t_0, x_0) \in \mathbb{R} \times \mathbb{R}^n$, let $\mathbf{u}$ be the control map given by Lemma 2.2. Then, from equation (13), it follows that: $V(t_0, x_0) \leq Q(t_0, x_0, \mathbf{u}) \leq \tilde{\alpha}_2(|x_0|_{\mathcal{A}}) \leq \alpha_2(|x_0|_{\mathcal{A}})$.

Lemma 2.5 There exists a $\mathcal{K}_\infty$ -map $\alpha_1(\cdot)$ such that, for all $(t_0, x_0) \in \mathbb{R} \times \mathbb{R}^n$, it holds that: $\alpha_1(|x_0|_{\mathcal{A}}) \leq V(t_0, x_0)$.

Proof. Fix any $(t_0, x_0) \in \mathbb{R} \times \mathbb{R}^n$, and let $i \geq 0$ be the first index such that $|x_0|_{\mathcal{A}} < \epsilon_i$. Let $\mathbf{w}$ be any control function such that the corresponding trajectory is defined for all $t \geq t_0$ and satisfies $|x(t)|_{\mathcal{A}} \leq \epsilon_{i+2}$. First, assume that $\|\mathbf{w}\| \leq \epsilon_{i+2}$, and let: $\delta_{x_0} = \frac{|x_0|_{\mathcal{A}}}{2}$. *Fact.* If $t \in [t_0, t_0 + \delta_{x_0})$, then $|x(t)|_{\mathcal{A}} > |x_0|_{\mathcal{A}}/2$.

Proof of Fact. If the conclusion does not hold, then, by continuity, there exists a $\bar{t} \in [t_0, t_0 + \delta_{x_0})$, such that $|x(\bar{t})|_{\mathcal{A}} = |x_0|_{\mathcal{A}}/2$. However, we have: $|x(\bar{t}) - x_0| \leq M_i(\bar{t} - t_0) < M_i\delta_{x_0} = |x_0|_{\mathcal{A}}/2$, which implies: $|x_0|_{\mathcal{A}} \leq |x(\bar{t})|_{\mathcal{A}} + |x(\bar{t}) - x_0| < \frac{|x_0|_{\mathcal{A}}}{2} + \frac{|x_0|_{\mathcal{A}}}{2} = |x_0|_{\mathcal{A}}$. So the fact is established. Now we have: $Q(t_0, x_0, \mathbf{w}) \geq \int_{t_0}^{t_0+\delta_{x_0}} M(|x(t)|_{\mathcal{A}})N(|x(t)|_{\mathcal{A}}) dt M(|x_0|_{\mathcal{A}}/2)N(|x_0|_{\mathcal{A}}/2) \frac{|x_0|_{\mathcal{A}}}{2M_i}$. Next, by combining equation (7) with the previous inequality, one gets: $Q(t_0, x_0, \mathbf{w}) \geq \frac{|x_0|_{\mathcal{A}}}{2} N(|x_0|_{\mathcal{A}}/2)$. If $\|\mathbf{w}\| > \epsilon_{i+2}$, then: $Q(t_0, x_0, \mathbf{w}) \geq \delta(\gamma(\epsilon_{i+2})) - \delta(\gamma(\epsilon_0)) \geq \frac{\epsilon_i}{2} N(\epsilon_i/2)$, where to get this last inequality, we have used inequality (9). So, in any case, we conclude: $V(t_0, x_0) \geq \frac{|x_0|_{\mathcal{A}}}{2} N(|x_0|_{\mathcal{A}}/2)$. Thus, by letting $\alpha_1(p) = p/2N(p/2)$, the conclusion follows.

Thus the function $V(\cdot, \cdot)$ satisfies property 1 of Definition 1.3. It remains to show that V is continuous and it satisfies also property 2 of the same definition. First, we prove continuity at the points $(t_0, x_0) \in \mathbb{R} \times \mathcal{A}$.

Lemma 2.6 The function $V(\cdot, \cdot)$ is continuous at $(t_0, x_0) \in \mathbb{R} \times \mathcal{A}$. Moreover it holds that for each $\epsilon > 0$ there exists a $\delta > 0$ such that if $|x|_{\mathcal{A}} \leq \delta$ then

$$|V(t, x)| \leq \epsilon \quad \forall t \in \mathbb{R}. \quad (14)$$

Proof. It is clear that it suffices to prove (14), since $V(t_0, x_0) = 0$ for all $(t_0, x_0) \in \mathbb{R} \times \mathcal{A}$. Given $\epsilon > 0$, choose $0 < \delta < \epsilon_0$ such that $M(\beta(p, 0))\beta(p, 0) \leq \epsilon$ for all $p \leq \delta$. Now if $|x|_{\mathcal{A}} \leq \delta$, then by Remark 2.3, property (a), we have: $V(t, x) \leq M(\beta(|x|_{\mathcal{A}}, 0))\beta(|x|_{\mathcal{A}}, 0) \leq \epsilon$; as desired.

Lemma 2.7 For each $(t_0, x_0) \in \mathbb{R} \times \mathbb{R}^n$, let $i \geq 0$ be the first index such that $|x_0|_{\mathcal{A}} < \epsilon_i$, and let $\mathbf{u}_0$ be the control map given by Lemma 2.2. If $\mathbf{w}$ is any control such that: $Q(t_0, x_0, \mathbf{w}) \leq Q(t_0, x_0, \mathbf{u}_0)$, then $\|\mathbf{w}\| \leq \epsilon_{i+2}$, and the corresponding trajectory $x(t)$ is such that:

$$|x(t)|_{\mathcal{A}} < \epsilon_{i+2}. \quad (15)$$

Proof. Assume that $\|\mathbf{w}\| > \epsilon_{i+2}$. Then it holds that: $Q(t_0, x_0, \mathbf{w}) \geq \delta(\|\mathbf{w}\|) - \delta(\gamma(\epsilon_0)) \geq \delta(\epsilon_{i+2}) - \delta(\gamma(\epsilon_0)) > F(\beta(\epsilon_i, 0))T_i + M(\beta(\epsilon_i, 0))\beta(\epsilon_i, 0) + \max\{\delta(\gamma(\epsilon_i)) - \delta(\gamma(\epsilon_0)), 0\}$, where, to get the last inequality, we have used (10). Notice that this last inequality contradicts the assumption $Q(t_0, x_0, \mathbf{w}) \leq Q(t_0, x_0, \mathbf{u}_0)$. Now, assume that $\|\mathbf{w}\| \leq \epsilon_{i+2}$, but that the inequality (15) is not satisfied. Then there exists $T > t_0$ such that $|x(t)|_{\mathcal{A}} = \epsilon_{i+2}$. Let

$$\tilde{T} = T + \frac{\epsilon_{i+2} - \epsilon_{i+1}}{2M_i}.$$

Then, for all $t \in [T, \tilde{T}]$, we have: $|x(t) - x(T)| \leq M_i(t - T) \leq 1/2(\epsilon_{i+2} - \epsilon_{i+1})$, from which it follows that $|x(t)|_{\mathcal{A}} \geq \epsilon_{i+1}$. So, we have: $Q(t_0, x_0, \mathbf{w}) \geq \int_T^{\tilde{T}} F(|x(t)|_{\mathcal{A}})dt \geq M(\epsilon_{i+1})N(\epsilon_{i+1})\frac{\epsilon_{i+2} - \epsilon_{i+1}}{2M_i} \geq F(\beta(\epsilon_i, 0))T_i + M(\beta(\epsilon_i, 0))\beta(\epsilon_i, 0) + \delta(\gamma(\epsilon_i))$, where the last inequality holds be equation (11). Thus, the assumption that $Q(t_0, x_0, \mathbf{w}) \leq Q(t_0, x_0, \mathbf{u}_0)$ is again contradicted.

Lemma 2.8 Let $t_n \rightarrow t_0$, $x_n \rightarrow x_0$, and $\mathbf{w}_n \rightarrow \mathbf{w}_0$, be such that $\|\mathbf{w}_n\| \leq r_1$, and all the solutions $x_n(t) = x(t; t_n, x_n, \mathbf{w}_n)$ exist for all $t \geq t_n$ and satisfy $|x_n(t)|_{\mathcal{A}} \leq r_2$, where r_1, r_2 are two positive real constants. Then the solution $x(t) = x(t; t_0, x_0, \mathbf{w}_0)$ exists for all $t \geq t_0$, and it is such that: $x_n(t) \rightarrow x(t)$ (as $n \rightarrow \infty$), and $|x(t)|_{\mathcal{A}} \leq r_2$, again for all $t \geq t_0$.

Proof. Since $x_n \rightarrow x_0$, we have $|x_0|_{\mathcal{A}} \leq r_2$. Given $\mathbf{w}_0$, by local existence of solutions, there exists an interval $[t_0, \bar{t}]$, in which $x(t)$ is defined. Since $x_n \rightarrow x_0$, we may assume that $x_n \in \text{clos } B(x_0, 1)$, and $t_n \in [t_0 - 1, t_0 + 1]$ for all $n \geq 1$. For each $t > t_0$, we let:

$$A(t) = \{x_n(\tau) \mid \tau \in [t_n, t]\} \subset \mathbb{R}^n.$$

Then $A(t)$ is bounded, in fact,:

$$|x_n(\tau)| \leq |x_n| + \int_{t_n}^{\tau} |f(s, x_n(s), u(s))|ds \leq$$

$$\leq |x_n| + M_{r_1 r_2}(\tau - t) \leq |x_0| + 1 + M_{r_1 r_2}\delta(t),$$

where $M_{r_1 r_2} = \sup |f(t, x, u)|$ for $|x|_{\mathcal{A}} \leq r_2$, and $\|u\| \leq r_1$, and $\delta(t) = (t - t_0 + 1)$. First we prove the following fact.

Fact. If $x(t)$ exists for all $t \in [t_0, t']$, then $x_n(t) \rightarrow x(t)$ as $n \rightarrow \infty$ for all $t \in [t_0, t']$.

Proof of Fact. Let:

$$K = \text{clos } A(t') \cup \{x(t) \mid t \in [t_0, t']\},$$

then K is a compact set. Let L_{K, r_1} , and $\tilde{M}$ be the Lipschitz constant and a bound for f with $x \in K$, $\|u\| \leq r_1$, and $t \in [t_0 - 1, t']$. Then one gets: $|x_n(t) - x(t)| \leq |x_n - x_0| + \tilde{M}|t_n - t_0| + L_{K, r_1} \int_{t_0 \vee t_n}^{t'} \|\mathbf{w}_n(s) - \mathbf{w}_0(s)\| + |x_n(s) - x(s)|ds$. Using the Bellman-Gronwall inequality, one gets: $|x_n(t) - x(t)| \leq (|x_n - x_0| + \tilde{M}|t_n - t_0| + L_{K, r_1} \int_{t_0 \vee t_n}^{t'} \|\mathbf{w}_n(s) - \mathbf{w}_0(s)\| ds) e^{L_{K, r_1}(t' - t_0 \vee t_n)}$. From this last inequality the fact easily follows.

Now we prove that $x(t)$ exists for all $t \geq t_0$. Assume that $x(t)$ exists only for $t \in [t_0, \bar{t}]$. Since, for $t \in [t_0, \bar{t}]$, we have $x_n(t) \rightarrow x(t)$ (from the previous fact), it holds that: $x(t) \in \text{clos } A(\bar{t})$. Since $\text{clos } A(\bar{t})$ is a compact set, this contradicts the fact that $x(t)$ does not exist for $t = \bar{t}$.

Lemma 2.9 Let $(t_n, x_n) \rightarrow (t_0, x_0)$, and $\mathbf{w}_n \rightarrow \mathbf{w}_0$. Let $i \geq 0$ be the first index such that $|x_0|_{\mathcal{A}} < \epsilon_i$. Assume that all $Q(t_n, x_n, \mathbf{w}_n)$ are finite, and that $\|\mathbf{w}_n\| \leq \epsilon_{i+2}$. Then:

$$Q(t_0, x_0, \mathbf{w}_0) \leq \liminf_{n \rightarrow +\infty} Q(t_n, x_n, \mathbf{w}_n).$$

Proof. For n sufficiently large, we must have $|x_n|_{\mathcal{A}} \leq \epsilon_i$, and $|x_n(t)|_{\mathcal{A}} \leq \epsilon_{i+2}$. Thus, from Lemma 2.8, it holds that $x_0(t)$ is defined for all $t \geq t_0$ and $|x(t)|_{\mathcal{A}} \leq \epsilon_{i+2}$. Moreover it is also true that: $F(|x_n(t)|_{\mathcal{A}}) \rightarrow F(|x_0(t)|_{\mathcal{A}})$. Given $\epsilon > 0$ there exists $\delta > 0$ such that: $\int_{t_0}^{t_0 + \delta} F(|x_0(t)|_{\mathcal{A}})dt < \epsilon$. By Fatou's Lemma, we get (notice that for n large $t_n \leq t_0 + \delta$):

$$\int_{t_0 + \delta}^{\infty} F(|x_0(t)|_{\mathcal{A}})dt \leq \liminf_{n \rightarrow \infty} \int_{t_0 + \delta}^{\infty} F(|x_n(t)|_{\mathcal{A}})dt.$$

On the other hand, one easily sees that:

$$\begin{aligned} & \max\{\delta(\|\mathbf{w}_0\|) - \delta(\gamma(\epsilon_0)), 0\} \leq \\ & \liminf_{n \rightarrow +\infty} \max\{\delta(\|\mathbf{w}_n\|) - \delta(\gamma(\epsilon_0)), 0\}. \end{aligned}$$

Summing up, we conclude:

$$\begin{aligned} Q(t_0, x_0, \mathbf{w}_0) & \leq \epsilon + \int_{t_0 + \delta}^{\infty} F(|x_0(t)|_{\mathcal{A}})dt \\ & \quad + \max\{\delta(\|\mathbf{w}_0\|) - \delta(\gamma(\epsilon_0)), 0\}, \end{aligned}$$

which is upper bounded by

$$\begin{aligned} \epsilon & + \liminf_{n \rightarrow \infty} \int_{t_0 + \delta}^{\infty} F(|x_n(t)|_{\mathcal{A}})dt \\ & + \liminf_{n \rightarrow +\infty} \max\{\delta(\|\mathbf{w}_n\|) - \delta(\gamma(\epsilon_0)), 0\} \end{aligned}$$

and is itself bounded by $\epsilon + \liminf_{n \rightarrow +\infty} Q(t_n, x_n, \mathbf{w}_n)$. From which the conclusion follows since ϵ was arbitrary.

The arguments used in the proofs of next lemmas are similar to the one used in [8] to prove the corresponding results.

Lemma 2.10 For each $(t_0, x_0) \in \mathbb{R} \times \mathbb{R}^n$ there exists $\mathbf{w}_0$ such that: $V(t_0, x_0) = Q(t_0, x_0, \mathbf{w}_0)$.

Proof. Assume that $i \geq 0$ is the first index such that $|x_0|_{\mathcal{A}} < \epsilon_i$. Let $\mathbf{w}_n$ be a minimizing sequence, then, by Lemma 2.7, it must hold that $\|\mathbf{w}_n\| \leq \epsilon_{i+2}$. Thus, by sequential compactness of $\mathbf{W}_{\epsilon_{i+2}}$, we may assume that $\mathbf{w}_n \rightarrow \mathbf{w}_0$ (possibly extracting a subsequence). From Lemma 2.9, we have: $Q(t_0, x_0, \mathbf{w}_0) \leq \liminf_{n \rightarrow \infty} Q(t_0, x_0, \mathbf{w}_n) = V(t_0, x_0) \leq Q(t_0, x_0, \mathbf{w}_0)$.

Lemma 2.11 The function $V(\cdot, \cdot)$ is lower semicontinuous.

Proof. Let $(t_n, x_n) \rightarrow (t_0, x_0)$. Then if $i \geq 0$ is the first index such that $|x_0|_{\mathcal{A}} < \epsilon_i$, we may also assume that $|x_n|_{\mathcal{A}} < \epsilon_i$. Let $\mathbf{w}_n$ be such that $V(t_n, x_n) = Q(t_n, x_n, \mathbf{w}_n)$, then, by Lemma 2.7, $\|\mathbf{w}_n\| \leq \epsilon_{i+2}$. Since $\mathbf{W}_{\epsilon_{i+2}}$ is sequentially compact, we may assume that $\mathbf{w}_n \rightarrow \mathbf{w}_0$. By applying Lemma 2.8, we know that the trajectory $x_0(t) = x(t; t_0, x_0, \mathbf{w}_0)$ exists for all $t \geq t_0$ and it is the limiting trajectory. Moreover by Lemma 2.9, we have: $V(t_0, x_0) \leq Q(t_0, x_0, \mathbf{w}_0) \leq \liminf_{n \rightarrow \infty} Q(t_n, x_n, \mathbf{w}_n) = \liminf_{n \rightarrow \infty} V(t_n, x_n)$.

Lemma 2.12 The function $V(\cdot, \cdot)$ is continuous.

Proof. We need only to prove upper semicontinuity. Fix $\epsilon > 0$, and let $0 < \delta < \epsilon_0$ be such that if $|x|_{\mathcal{A}} \leq \delta$ then $V(t, x) \leq \epsilon/3$ for all $t \in \mathbb{R}$ (use Lemma 2.6). Fix any $(t_0, x_0) \in \mathbb{R} \times \mathbb{R}^n$. Let $i \geq 0$ be the first index such that $|x_0|_{\mathcal{A}} < \epsilon_i$, and $\mathbf{w}_0$ be such that $V(t_0, x_0) = Q(t_0, x_0, \mathbf{w}_0)$. Then $\|\mathbf{w}_0\| \leq \epsilon_{i+2}$ (Lemma 2.7). Denote by $x(t) = x(t; t_0, x_0, \mathbf{w}_0)$, then, again by Lemma 2.7, we know that $|x(t)|_{\mathcal{A}} < \epsilon_{i+2}$, and that there exists T such that $x(T) < \delta$. By continuity, there exist a neighborhood $J \times H$ of (t_0, x_0) such that, for all $(t', z) \in J \times H$, $|z|_{\mathcal{A}} < \epsilon_i$, $z(t) = x(t; t', z, \mathbf{w}_0)$ exists for all $t \in [t', T]$, $|z(t)|_{\mathcal{A}} < \epsilon_{i+2}$, $|z(T)|_{\mathcal{A}} < \delta$, and

$$\int_{t'}^T F(|z(t)|_{\mathcal{A}}) < \int_{t'}^T F(|x(t)|_{\mathcal{A}}) + \epsilon/3.$$

Now we have:

$$\begin{aligned} V(t', z) &\leq \epsilon/3 + \int_{t'}^T F(|z(t)|_{\mathcal{A}}) + \\ &\max\{\delta(\|\mathbf{w}_0\|) - \delta(\gamma(\epsilon_0)), 0\} < \int_{t_0}^T F(|x(t)|_{\mathcal{A}}) + \\ &\frac{2\epsilon}{3} + \max\{\delta(\|\mathbf{w}_0\|) - \delta(\gamma(\epsilon_0)), 0\} \leq V(t_0, x_0) + \epsilon. \end{aligned}$$

Thus V is upper semicontinuous.

Lemma 2.13 The function $V(\cdot, \cdot)$ is a Lyapunov function.

Proof. We have already shown that V is continuous and satisfies property 1. of Definition 1.3. We need only to prove that also property 2. holds. Fix any $(t_0, x_0) \in \mathbb{R} \times \mathbb{R}^n$. Let $i \geq 0$ be the first index such that $|x_0|_{\mathcal{A}} < \epsilon_i$, and $\mathbf{w}_0$ be such that

$V(t_0, x_0) = Q(t_0, x_0, \mathbf{w}_0)$. Let γ_1 be any continuous, positive and increasing map such that $\gamma_1(p) \geq \epsilon_{i+2}$ for $\epsilon_{i-1} \leq p < \epsilon_i$, for $i \geq 0$ (set $\epsilon_{-1} = 0$). Denote by $x(t) = x(t; t_0, x_0, \mathbf{w}_0)$, then we know that $|x(t)|_{\mathcal{A}} < \epsilon_{i+2}$. For t sufficiently close to t_0 we have that $|x(t)|_{\mathcal{A}} < \epsilon_i$. Moreover, if we denote by $\mathbf{w}'_0$ the translation of $\mathbf{w}_0$ by $(-t)$, we have:

$$\begin{aligned} V(t, x(t)) &\leq \int_t^\infty F(|x(s)|_{\mathcal{A}}) ds + \\ &+ \max\{\delta(\|\mathbf{w}'_0\|) - \delta(\gamma(\epsilon_0)), 0\}, \end{aligned}$$

thus (notice that $\|\mathbf{w}'_0\| \leq \|\mathbf{w}_0\|$):

$$\begin{aligned} \liminf_{t \rightarrow t_0^+} \frac{V(t, x(t)) - V(t_0, x_0)}{t - t_0} &\leq \\ \liminf_{t \rightarrow t_0^+} \frac{1}{t - t_0} \left(- \int_{t_0}^t F(|x(s)|_{\mathcal{A}}) ds \right) &= -F(|x_0|_{\mathcal{A}}). \end{aligned}$$

So also property 2. holds with $\alpha_3 = F$, and $\phi = \gamma_1$.

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